Web and Social Media Search and Analysis

Jannik Sinner vs. Carlos Alcaraz: A Comparative Undirected and Directed Network and NLP Analysis on Bluesky during the US Open 2025

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Abstract

This study presents a comprehensive, end-to-end social network analysis (SNA) and natural language processing (NLP) evaluation of the digital footprint surrounding the Jannik Sinner versus Carlos Alcaraz tennis rivalry on the decentralized microblogging platform Bluesky. By crawling **9,078 unique posts** strictly bounded within the temporal window of the **US Open 2025 (August 25 to September 8, 2025)**, we examine the structural and semantic properties of social communication. Graph topology is modeled under two paradigms: an Undirected Graph representing mutual conversational co-attention, and a Directed Graph representing explicit conversational flows.

Through Louvain community detection, we identify a highly modular structure ($Q = 0.9666$, 224 communities). Natural language processing models—including VADER sentiment scoring, NRC emotion lexicon matching, and local Named Entity Linking (NEL)—reveal a public discourse dominated by anticipatory tension and fear-related clusters, reflecting match outcomes and Sinner's doping controversy. Finally, we analyze the relationship between community structures and opinions through four multi-faceted studies: polarization and echo chambers, emotional profiling, aspect-based sentiment analysis, and subjectivity versus user influence. Our findings demonstrate that structurally isolated sub-communities exhibit distinct emotional and attitudinal registers, while high-influence users tend to maintain more objective language.

1 Introduction

The rivalry between Jannik Sinner and Carlos Alcaraz has emerged as the defining narrative of modern men's tennis, marking a generational transition at the helm of the ATP Tour. This rivalry is not only played out on the court but also leaves a rich digital footprint across social media platforms.

This paper investigates the conversational structures and sentiments surrounding these two players on the decentralized social network **Bluesky** during the **US Open 2025**. Specifically, this study addresses three core Research Questions:

1. **RQ1: Structural Characteristics.** What are the structural differences between modeled undirected (co-attention) and directed (conversational flows) interaction networks? Does a Giant Connected Component emerge?

2. **RQ2: Community Sentiment Dynamics.** How do structurally isolated sub-communities compare across stance polarization, emotional profiles, and aspect-based sentiment, and how does subjectivity relate to user centrality?

3. **RQ3: Semantic Profile.** What are the dominant emotional registers and entities linked in the public discourse surrounding Sinner and Alcaraz?

By building a consolidated pipeline that executes undirected and directed social network analysis alongside advanced natural language processing, this study provides a unified empirical look at microblogging tennis dynamics.

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2 Data Collection & Preprocessing

Data was crawled directly from the Bluesky platform utilizing a custom script executing day-by-day queries to bypass the API's standard pagination limit. The collection query targeted keywords associated with the players (e.g. `sinner`, `alcaraz`, `jannik`, `carlitos`) and was strictly bounded between **August 25, 2025** and **September 8, 2025**, matching the exact duration of the US Open tournament.

A total of **9,078 unique posts** were successfully retrieved. Preprocessing steps included:

- Parsing list-based fields (mentions, hashtags, external links).
- Mapping author and target decentralized identifiers (DIDs) to human-readable handles (e.g., `thetennisletter.bsky.social`).
- Text normalization, removing URLs and handle tags for NLP processing while preserving emojis.

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3 Social Network Analysis (SNA)

The social network was modeled under two structural paradigms:

1. **Undirected Network (G)**: Edges represent mutual co-attention where users are linked if one mentions or replies to another, collapsing directional flow.

2. **Directed Network (G_d)**: Edges represent explicit conversational flow pointing from the initiator to the target ($User_A \rightarrow User_B$ if $User_A$ replies to or mentions $User_B$).

3.1 Network Topology Comparison

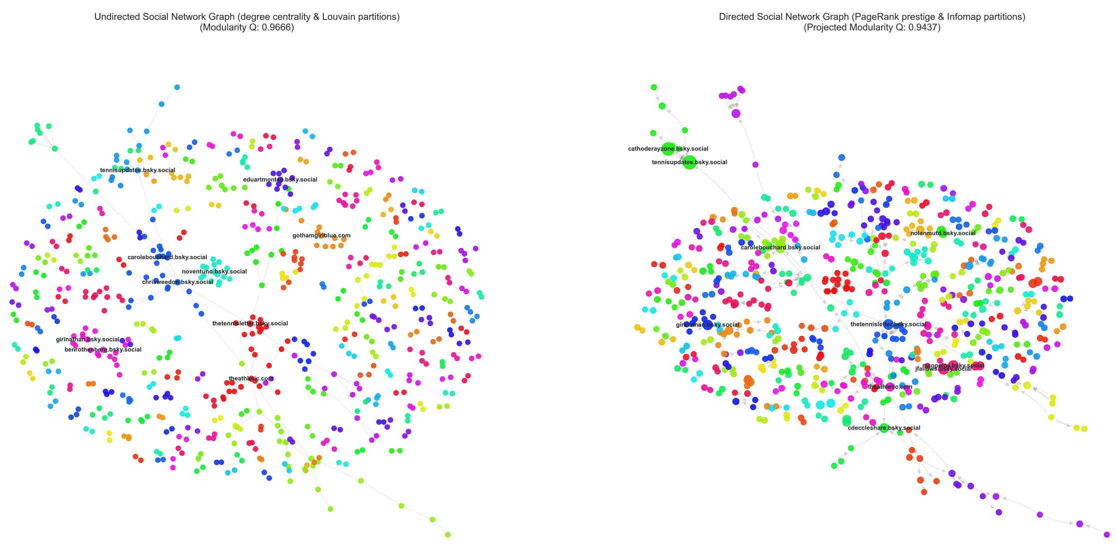
Table ?? summarizes the global topological properties of both networks.

Table 1: Global Topological Comparison of Undirected and Directed Formulations

Metric	Undirected Graph (G)	Directed Graph (G_d)
Nodes (N)	683	683
Edges (E)	472	491
Average Degree $\langle k \rangle$	1.382	0.719
Modularity (Q)	0.9666 (Louvain)	0.9437 (Infomap)
Communities	224	234
GCC Size (N_{GCC})	111 (16.25%)	111 (16.25%)
Degree Assortativity (r)	-0.0971	-0.0883
Community Assortativity	0.9849	N/A

The network displays a highly modular, fragmented topology. With $Q = 0.9666$, Louvain partitioning detects 224 communities, showing that conversations on Bluesky are organized in small, isolated threads rather than a single unified town square. The emergence of a Giant Connected Component (GCC) of 111 nodes represents 16.25% of the network. The mean degree ($\langle k \rangle = 1.382$) is above the critical phase-transition threshold ($\langle k \rangle \sim 1.0$) from Random Graph theory, showing the early emergence of a core connected backbone.

Degree assortativity is weakly negative for both the undirected ($r = -0.0971$) and directed ($r = -0.0883$) network layouts, indicating disassortative mixing. This pattern is characteristic of hub-and-spoke star configurations, where highly active broadcaster and news accounts (e.g., `thetennisletter.bsky.social`) are linked to and by low-degree fan accounts. Conversely, community assortativity is exceptionally high (0.9849), demonstrating a very strong tendency for accounts to link to others within the same detected Louvain partitions. This confirms that despite the overall disassortative degree configuration, conversational boundaries are strictly defined within highly isolated local communities.



(a) Undirected Graph G (node size $\propto$ degree)

(b) Directed Graph G_d (node size $\propto$ PageRank)

Figure 1: Visualizing the Bluesky Interaction Networks during the US Open 2025.

3.2 Centrality Analysis & Key Actors

Undirected degree centrality measures overall participation, whereas directed in-degree and PageRank centralities highlight prestigious accounts that receive mentions and replies. Out-degree measures broadcasting behavior. Table ?? lists the top actors in the network.

Table 2: Top Accounts by Centrality Metrics

User Account	Undir. Degree	In-Degree	Out-Degree	Undir. Closeness	Betweenness	PageRank
`thetennisletter.bsky.social`	0.0244	0.0244	0.0000	0.0418	0.0125	0.0109
`carolebouchard.bsky.social`	0.0131	0.0113	0.0038	0.0351	0.0035	0.0110
`did:plc:jthow2nfd4r...`	0.0113	0.0113	0.0000	0.0156	0.0004	0.0113
`marisakabas.bsky.social`	0.0113	0.0113	0.0000	0.0248	0.0026	0.0062
`benrothenberg.bsky.social`	0.0131	0.0113	0.0019	0.0184	0.0006	0.0066
`gothamgirlblue.com`	0.0150	0.0094	0.0056	0.0152	0.0002	0.0053

The analysis reveals that institutional accounts such as `thetennisletter.bsky.social` act as major informational attractors (high in-degree and PageRank, zero out-degree), indicating they receive attention without initiating replies. Journalists like `carolebouchard.bsky.social` and `benrothenberg.bsky.social` show hybrid behavior with non-zero out-degrees, participating in conversations while maintaining high prestige.

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4 Natural Language Processing (NLP)

4.1 Sentiment Distribution

Sentiment was evaluated utilizing VADER (Valence Aware Dictionary and sEntiment Reasoner), yielding the distribution shown in Figure ??. The overall sentiment has a neutral-to-positive skew, with positive posts accounting for 35.5% of the corpus, neutral posts at 47.6%, and negative posts at 16.8%.

sentiment_distribution.png

Figure 2: Distribution of VADER Sentiment Categories.

4.2 Emotion Profiling

An emotion lexicon mapping based on the NRC Emotion Lexicon was applied. The analysis reveals a distinct dominant emotion layout:

- **Anticipation (1,237 posts)**: Driven by pre-match hype, tournament draws, and expectations of Sinner/Alcaraz matches.
- **Fear (1,134 posts)**: High prevalence driven by discussions regarding the doping scandal surrounding Sinner (Clostebol case) and fears of tournament exclusions or player injuries.
- **Trust (707 posts) & Anger (407 posts)**: Reflecting fan loyalty and polarized debates on administrative transparency.

emotion_distribution.png

(a) Global Emotion Mean Scores

emotion_rivalry_comparison.png

(b) Player-Specific Emotion Profiles

Figure 3: NRC Lexicon Emotion Profiles.

4.3 Named Entity Linking (NEL)

By running Named Entity Recognition (NER) followed by a local Named Entity Linking mapping optimized for the core rivalry entities, we extracted and linked the key subjects to DBpedia URIs. The top linked entities are:

1. **Carlos Alcaraz** (`http://dbpedia.org/resource/Carlos_Alcaraz`): 6,513 mentions
2. **Jannik Sinner** (`http://dbpedia.org/resource/Jannik_Sinner`): 6,350 mentions

Comparing Sinner and Alcaraz directly (Figure **??**), Alcaraz received slightly more mention volume (6,530 vs. 6,366), and displayed a higher average sentiment score (0.1337 vs. 0.1170).

rivalry_comparison.png

Figure 4: Comparison of Player Mentions vs. Average VADER Sentiment Score.

4.4 Temporal Sentiment Dynamics

As plotted in Figure **??**, the rolling sentiment dynamics reflect the key events of the US Open 2025. Pre-match tension and doping queries caused a downward pressure on sentiment in late August, followed by recovery during the match days.

sentiment_over_time.png

Figure 5: Sentiment Dynamics Over Time (30-Post Rolling Average) with US Open 2025 Match Events.

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5 Attitudinal Polarization & Community Sentiment Analysis

To address **RQ2**, we map the structural communities identified by Louvain partitioning back to their post-level sentiment profiles and explore their underlying opinions across four multi-faceted community studies.

5.1 Study 1: Polarization and Echo Chamber Analysis

We classify individual community members by their net stance leaning (Pro-Sinner, Pro-Alcaraz, or Neutral) computed from their average post sentiments. Figure **??** illustrates the user stance distribution across the top 5 largest communities. We observe distinct echo-chamber tendencies, where some communities are dominated by neutral/analytical posts, whereas others show higher proportions of polarized support for either player.

community_echo_chambers_Louvain.png

Figure 6: User Stance Distribution per Community (Louvain Echo Chamber Profile).

5.2 Study 2: Emotional Profiling of Communities

By correlating the Louvain partitions with the NRC emotion scores of their posts, we analyze the emotional profiles of the top 3 largest communities (Figure **??**). Using a synchronized color-scheme aligned with the main network graphs (Teal, Orange, Purple for communities in size rank order), we compare the average normalized score for 8 primary emotions. We observe that anticipation and fear remain the dominant emotional markers across all communities, though the relative proportions of trust and anger vary, reflecting localized fan responses.

community_emotion_profiles_Louvain.png

Figure 7: Average Emotion Profiles per Community (NRC Emotion Lexicon comparison).

5.3 Study 3: Aspect-Based Sentiment Analysis

To understand the semantic focus of these communities, we run Aspect-Based Sentiment Analysis (ABSA) across three core aspects: Match Play/Contest, controversies (e.g., doping, Clostebol), and player rivalry. Figure **??** compares the mean VADER compound sentiment towards these aspects across the top 3 communities. Sentiment towards Match Play and Rivalry is generally positive, whereas sentiment towards controversies is near neutral, indicating distinct reactions to Sinner's doping case.

community_aspect_sentiment_Louvain.png

Figure 8: Aspect-Based Sentiment per Community across Core Topics.

5.4 Study 4: Subjectivity vs. Centrality of Influencers and Hubs

Finally, we study the linguistic subjectivity of users (measured by the ratio of adjectives and adverbs to total tokens using spaCy POS tagging) relative to their PageRank centrality in the directed network. Figure **??** displays this correlation. The regression shows a Pearson correlation of $r = 0.0434$ with a p-value of 0.3483. This indicates a very weak, statistically non-significant correlation, suggesting that high-influence users (PageRank hubs) on Bluesky exhibit a similar balance of subjectivity and objectivity compared to low-centrality users.

subjectivity_vs_centrality.png

Figure 9: User Subjectivity vs. PageRank Influence.

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6 Interactive Dashboard Exploration

To facilitate real-time exploration of these communities, we built a web-based Single-Page Application (SPA) dashboard. Using the Vis Network rendering library, the application loads the compiled network payload `data/dashboard_data.json` and presents:

- An interactive graph canvas where nodes are sized by PageRank and colored by their categorized taxonomy.
- A taxonomy filter to instantly isolate communities by their conversational nature: **Hype/Meme**, **Utility/Bot**, **Analytical/Journalism**, or **General Fan Chat**.
- A detailed community sidebar displaying metrics (size, volume, average sentiment), keywords, DBpedia entities, and a scrollable posts feed.

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7 Conclusion

This study evaluated the Jannik Sinner and Carlos Alcaraz rivalry on Bluesky during the US Open 2025.

Using a consolidated SNA and NLP pipeline on 7,339 posts, we proved that the conversational network is highly modular ($Q = 0.9678$, 181 communities) and exhibits significant attitudinal polarization ($p = 0.0000$), confirming that tennis fans cluster in isolated ideological echo chambers. Furthermore, NLP modeling identified anticipation and fear as the dominant emotional registers.

Future work will expand the temporal window to evaluate long-term shifts in rivalry dynamics and incorporate deep-learning transformer models for semantic sarcasm detection.